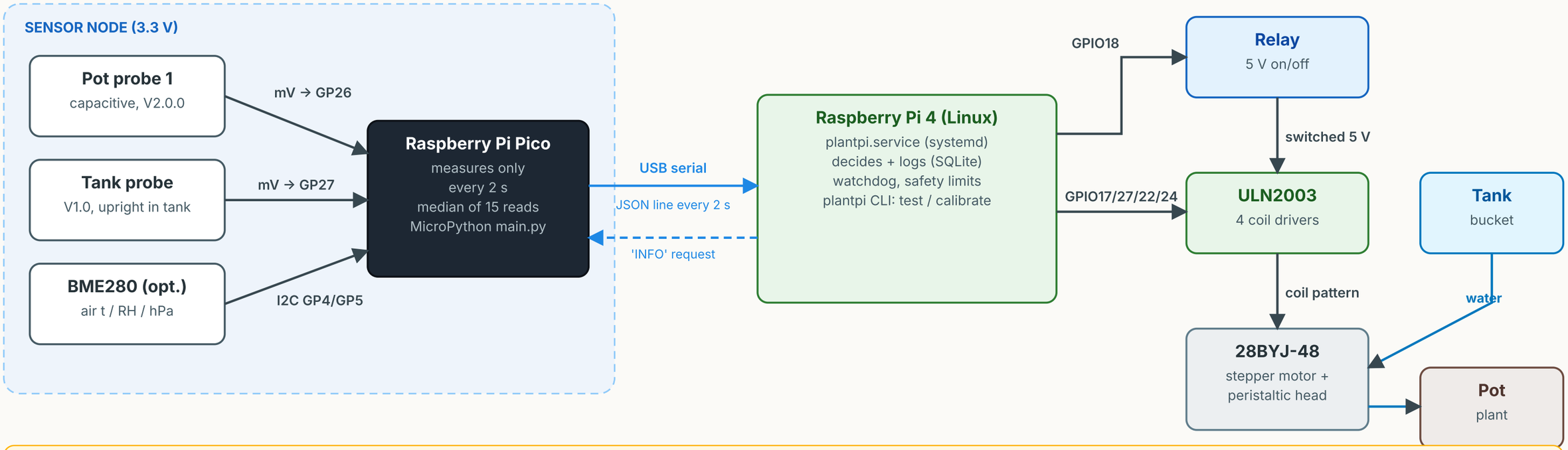
PlantPi - wiring, soldering and limits

Group 19 - Embedded Linux (EMLI), SDU. Raspberry Pi Pico sensor node + Raspberry Pi 4 + stepper peristaltic pump. One pot now, Pot 2 prepared. Wire numbers match `plantpi pins` and the code on GitHub.

Three rules before you plug anything in.

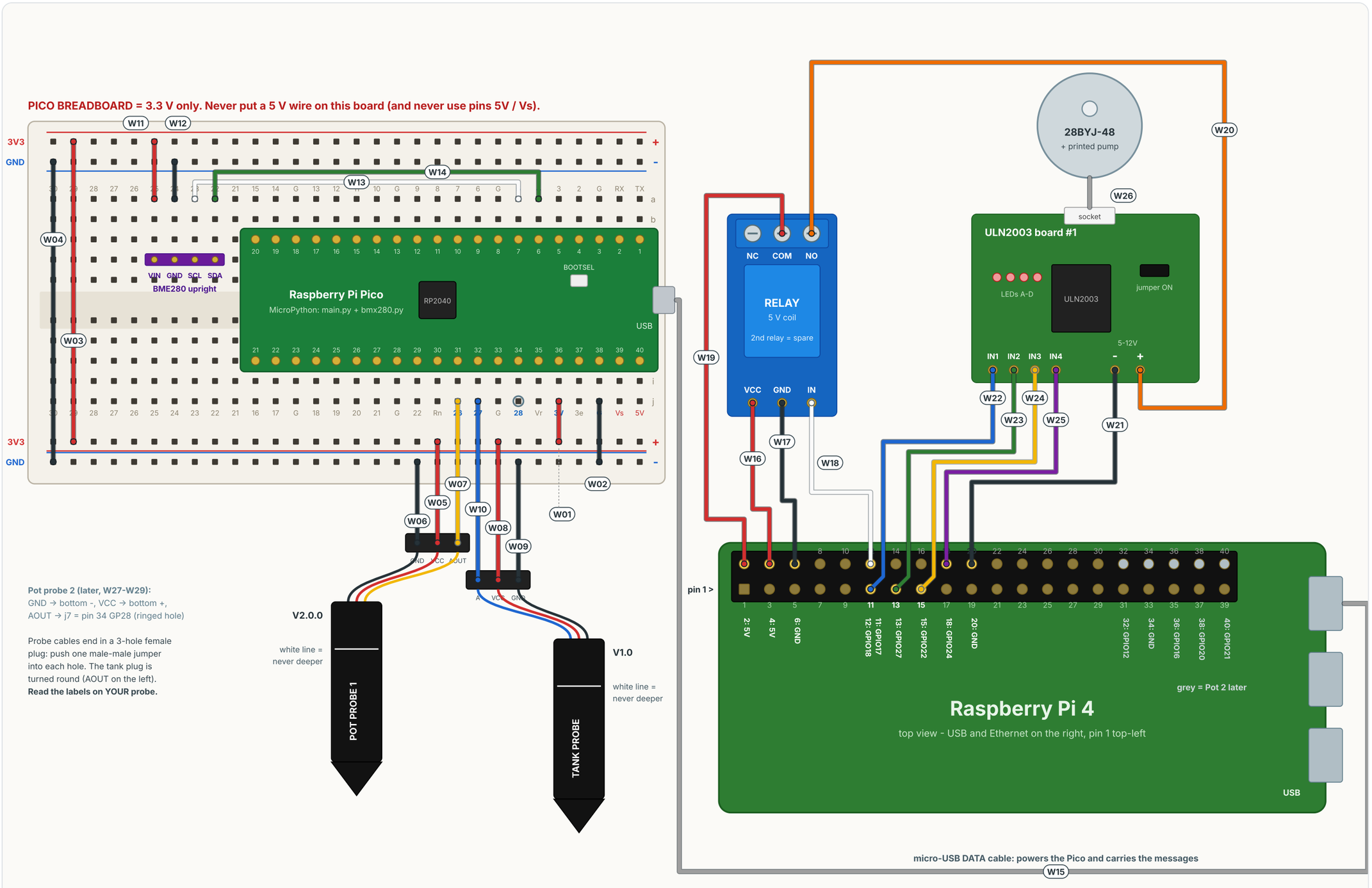
1. The Pico breadboard is **3.3 V only**. No wire from a Pi 5 V pin goes onto it, and the Pico pins **5V** and **Vs** stay unused.
2. Wire with the Pi **switched off** and the Pico **unplugged**. Check each wire against the table, then power up.
3. Do not use the LiPo battery, the 12 V supply or the DC-jack adapters: nothing here needs them.

1. How it fits together



The Pico only measures and reports; the Pi decides, logs and drives the pump. Adding a sensor means one line in the firmware and one name in the config.

2. Build view: what goes where



Drawn like your kit's Pico breadboard: USB at the right end, column 1 on the right, and the same pin labels that are printed on the board (3V, G, 26, 27, 28, 4, 5). Small numbers on the Pico = its physical pin numbers. Tags W01-W26 = wiring table below; dashed = later (Pot 2).

Pico on its breadboard

Leave the Pico where it is (kit rule). Its pins sit in rows **c** and **h**, so rows a-b and i-j next to every pin are free for jumpers. Use the labels printed on the board: **3V** = pin 36, **26 / 27 / 28** = analog GP26 / GP27 / GP28, top-row **4 / 5** = GP4 (SDA) / GP5 (SCL).

Pi side has no breadboard

Relay and ULN2003 pins are male, the Pi header is male: use **female-female** jumpers. Screw terminals (COM, NO) take the **male** end of a jumper: loosen, insert, tighten, pull gently to check.

Motor plug

The 5-wire white plug of the 28BYJ-48 fits the ULN2003 socket only one way. Keep the small **jumper** on the ULN2003 board in place (it connects the motor power).

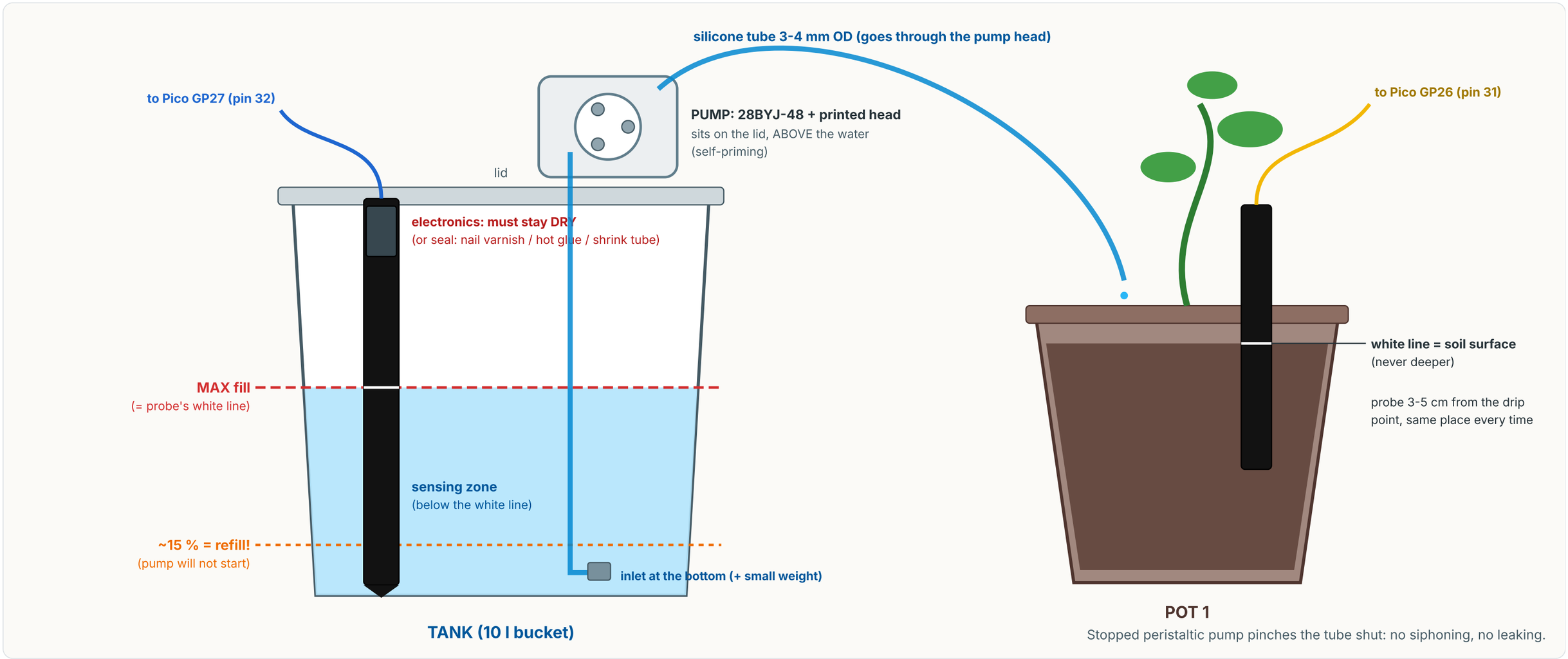
3. Wiring table

#	From	To	Colour	Jumper
Now: pot 1				
W01	Pico pin 36 3V3 (hole j5)	bottom + rail	red	M-M
W02	Pico pin 38 GND (hole j3)	bottom - rail	black	M-M
W03	bottom + rail	top + rail (bridge)	red	M-M
W04	bottom - rail	top - rail (bridge)	black	M-M
W05	bottom + rail	pot probe 1 VCC	red	M-M into probe cable
W06	bottom - rail	pot probe 1 GND	black	M-M into probe cable
W07	pot probe 1 AOUT	Pico pin 31 GP26/ADC0 (hole j10)	yellow	M-M into probe cable
W08	bottom + rail	tank probe VCC (+)	red	M-M into probe cable
W09	bottom - rail	tank probe GND (-)	black	M-M into probe cable
W10	tank probe AOUT (A)	Pico pin 32 GP27/ADC1 (hole j9)	blue	M-M into probe cable
W11	top + rail	BME280 VIN	red	M-M
W12	top - rail	BME280 GND	black	M-M
W13	BME280 SCL	Pico pin 7 GP5 (hole a7)	white	M-M
W14	BME280 SDA	Pico pin 6 GP4 (hole a6)	green	M-M
W15	Pico micro-USB socket	any Pi USB port	grey	USB data cable
W16	Pi pin 4 (5V)	relay VCC	red	F-F
W17	Pi pin 6 (GND)	relay GND	black	F-F
W18	Pi pin 12 (GPIO18)	relay IN	white	F-F
W19	Pi pin 2 (5V)	relay COM (screw)	red	F-M
W20	relay NO (screw)	ULN2003 #1 +	orange	M-F
W21	Pi pin 20 (GND)	ULN2003 #1 -	black	F-F
W22	Pi pin 11 (GPIO17)	ULN2003 #1 IN1	blue	F-F
W23	Pi pin 13 (GPIO27)	ULN2003 #1 IN2	green	F-F
W24	Pi pin 15 (GPIO22)	ULN2003 #1 IN3	yellow	F-F
W25	Pi pin 18 (GPIO24)	ULN2003 #1 IN4	purple	F-F

#	From	To	Colour	Jumper
W26	28BYJ-48 #1 plug	ULN2003 #1 white socket	grey	motor cable
Later: pot 2				
W27	bottom + rail	pot probe 2 VCC	red	M-M into probe cable
W28	bottom - rail	pot probe 2 GND	black	M-M into probe cable
W29	pot probe 2 AOUT	Pico pin 34 GP28/ADC2 (hole j7)	yellow	M-M into probe cable
W30	relay NO (same screw as W20)	ULN2003 #2 +	orange	M-F
W31	Pi pin 34 (GND)	ULN2003 #2 -	black	F-F
W32	Pi pin 32 (GPIO12)	ULN2003 #2 IN1	blue	F-F
W33	Pi pin 36 (GPIO16)	ULN2003 #2 IN2	green	F-F
W34	Pi pin 38 (GPIO20)	ULN2003 #2 IN3	yellow	F-F
W35	Pi pin 40 (GPIO21)	ULN2003 #2 IN4	purple	F-F
W36	28BYJ-48 #2 plug	ULN2003 #2 white socket	grey	motor cable

F-F = female-female jumper, M-M = male-male, F-M = female on one end, male on the other.

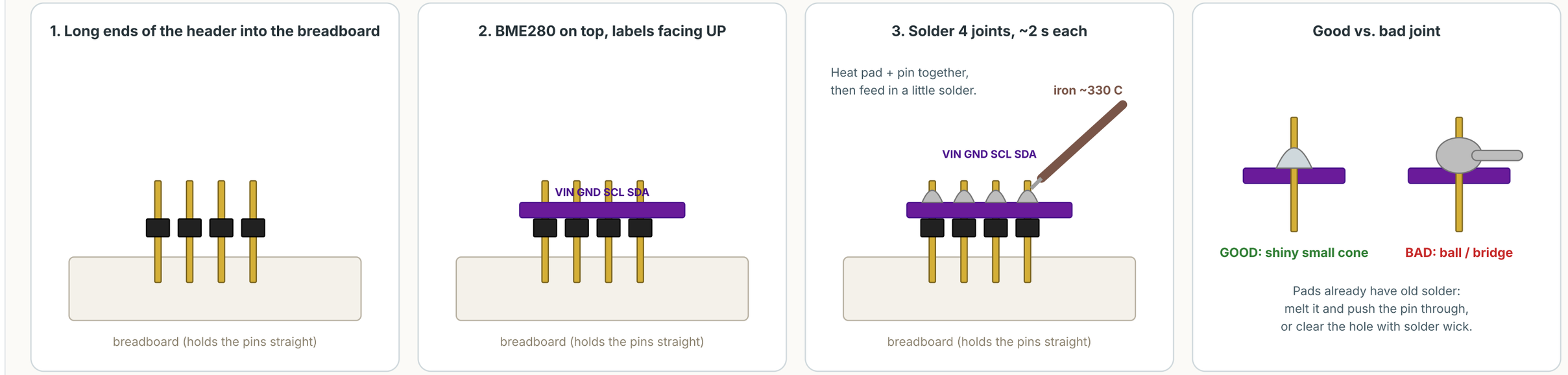
4. Tank, pump and pot



The tank probe is the V1.0 probe standing upright, tip at the bottom. Its reading falls as the water rises, but only along the sensing zone below its white line.

5. What to solder

Only one thing: the 4-pin header of the BME280/BMP280 air sensor (4 joints). Everything else plugs in or screws in. The kit rules say no soldering, so ask the lab/TA for the soldering station first. **The project works without the air sensor:** skip it if soldering is not allowed; the firmware simply sends no air data.



Heat the pad and the pin together for 1-2 s, then add a little solder. Let it cool without moving. Check that no solder joins two neighbouring pins.

Optional, only if a test fails: on some cheap V2.0 soil probes the reading never drops in water, or drops very slowly, because a discharge resistor is not connected to ground. `plantpi test node` shows it. The fix is one short wire soldered on the probe; otherwise use the other probe.

6. Limits you should know

Pump (28BYJ-48)

- **Slow and weak:** about 10 rpm and little torque. A peristaltic pump on it moves only a few ml per minute, so 30 ml takes about 10 min. Measure it: `plantpi test pump 1`.
- Needs **soft silicone tube** (3 or 4 mm outside diameter, matching the printed head). The 8 mm PVC hose cannot go through the pump head.
- Rollers too tight: the motor stalls and buzzes. Too loose: no flow. The motor has no sensor, so the software cannot see a stall: the ml are an estimate.
- One run is capped at 20 min / 250 ml. The motor gets warm on long runs; the coils are switched off after each run.
- No samples are logged while the pump runs, but the tank and the node are still checked every 0.5 s.

Tank level (moisture probe)

- Coarse: it only "sees" the bottom part of the tank (the zone below its white line). It is a **low-water alarm**, not a gauge. Fill no higher than the MAX line.
- Electronics at the top are **not waterproof**. Splashes kill it; seal it if you can.
- Drops on the probe make the level lag a little, so the code starts at 15 % and stops below 10 %.

Soil probes

- Values depend on the soil, how hard it is pressed and the temperature. Calibration in air and water gives the scale; the 35 % threshold has to be tuned by watching the plant.
- V1.0 and V2.0.0 read differently: each probe is calibrated on its own (`plantpi calibrate 1`, `plantpi calibrate tank`).
- Same insertion depth and position every time, or the numbers jump.

Raspberry Pi Pico

- 3 analog inputs (GP26, GP27, GP28), 12-bit, 0-3.3 V. The probes run on the Pico's 3.3 V, so their signal can never go above that.
- The RP2040 ADC has a few known odd codes and some noise; the median of 15 reads removes them. Fine for probes.
- The Feather HUZZAH (ESP8266) cannot do this job: it has one analog pin, 0-1 V. It stays free for Wi-Fi later (Block 2).
- Use a micro-USB **data** cable. The Pico only sends while the Pi has the port open; the Pi asks it to introduce itself ("INFO") every time it connects.
- If the Pico is silent for 10 s, the pump stops and nothing is watered until data is back.

Power and relay

- Everything runs from the Pi's 5.1 V 3 A supply: motor ~0.25 A, relay ~0.07 A, Pico ~0.03 A. Only one motor runs at a time. Never power the Pi from a laptop port.
- It is not known if the relay modules switch on HIGH or LOW: `plantpi test relay 1` tells you, and `relay_active_high` in the config fixes it.
- One relay powers both motor drivers, so the second relay module is a spare.

System

- One pot now. Pot 2 needs one more 28BYJ-48 + ULN2003 board; probe, relay and code are ready.
- The Pi has no clock battery: times are marked unsure until the Pi has synced over the internet.
- The air sensor is labelled "BME/BMP280": the firmware detects which. A BMP280 has no humidity.
- USB cable now. Wi-Fi / MQTT is a later step: same messages, new transport.

7. Build order (one check after each step)

1. Pico: hold **BOOTSEL**, plug it into a laptop and drag the MicroPython .uf2 file onto the RPI-RP2 drive. In Thonny, save `main.py` and `bmx280.py` onto the Pico. Thonny's Shell then prints a `{"v":1,...}` line every 2 s.
2. Pico breadboard: rails W01-W04, pot probe 1 W05-W07, tank probe W08-W10, (BME280 W11-W14). Touch a probe and watch its number change in Thonny.
3. Pico USB into the Pi (W15). On the Pi: `plantpi test node`.
4. Pi off. Relay W16-W19. Pi on: `plantpi test relay 1`.
5. Pi off. ULN2003 W20-W25, motor plug W26. Pi on: `plantpi test motor 1`.
6. Pump head on the motor, inlet tube in the tank, outlet in a measuring cup: `plantpi test pump 1`, copy `ml_per_rev` into `/etc/plantpi/config.toml`.
7. `plantpi calibrate 1` and `plantpi calibrate tank`, then `sudo systemctl enable --now plantpi`.